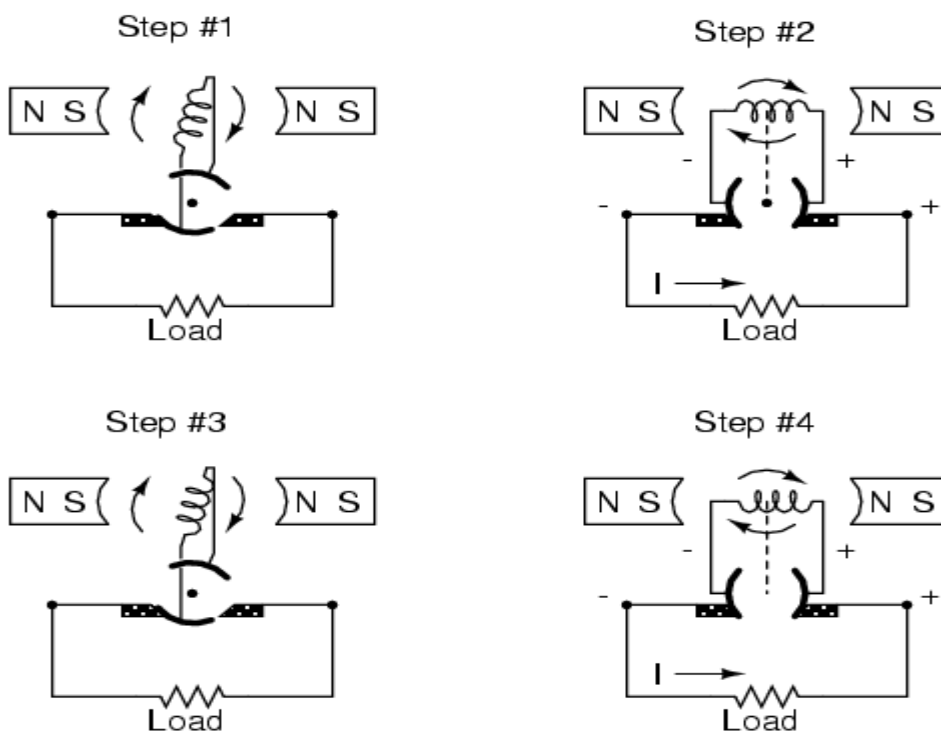


UNIT- IIID.C Generator

Principle of operation

A d-c generator converts mechanical energy into electrical energy. It works on Faraday's laws of electromagnetic induction principle. According to this law, whenever a conductor cuts the magnetic flux, dynamically induced emf is produced in the conductor. The direction of the emf induced is given by Fleming's Right Hand Rule. This emf causes a current to flow when the conductor circuit is closed. Thus, the essential parts of the generator are (i) a conductor and (ii) a magnetic field.

The figure 1 shows the arrangement of a single turn coil rotating at a constant speed in an uniform magnetic field produced by two poles. As the coil rotates, it takes different positions in the magnetic field. Since the flux linking with the coil changes with position of the coil, the magnitude of emf induced in the coil also changes with the position of the coil as shown in Figure 2.



At the positions '1' and '3' of the coil, the rate of change of flux linkages is minimum and hence the emf induced is also minimum. However, at position '2' the rate of change of flux linkages is maximum and hence the emf induced is also maximum. This process repeats in the next half cycle. From the Figure 2 it is clear that, even in the d.c. generator, the emf

induced in the coil (armature) is alternating. A commutator is used to convert this alternating voltage induced in the coil (armature) into uni-directional current in the external circuit.

Constructional Details

The D.C. generator consists of the following parts.

- (I) Yoke or magnetic-frame
- (II) Pole core and pole shoe
- (III) Pole coil or field coil or Exciting coil
- (IV) Armature core
- (V) Armature winding
- (VI) Commutator
- (VII) Brushes
- (VIII) Bearings.

Yoke

The purpose of the yoke is to provide support to the armature and to the magnetic field windings. It carries the magnetic flux produced by the poles. It acts as a protecting cover to the entire machine. The terminal box, lifting eye and the feet are welded to the frame as shown in Figure 3. For small size a generator, cast-iron is used where as for medium and large size machines cast steel or rolled steel is used for the yoke.

Pole core and poles shoe

The pole core is made of thin laminations of annealed steel of thickness 1 mm to 0.25 mm. The outer end of the poles core is curved to fit very closely into the inner surface of the yoke as shown in Figure 4(a). The laminations are riveted together under hydraulic pressure. The pole core is fixed to the yoke by means of a screw bolted through the yoke into the body of the pole core as shown in Figure 4(b)

The pole shoe acts as a support to the pole coil. It spreads the flux in the air gap. Being larger in cross section, it reduces the reluctance of the magnetic path.

Pole coil

The pole coil consists of copper wires which are 'former wound' for correct dimension as shown in Figure 5. Whenever they carry current, they electro magnetize the poles which produced necessary magnetic flux to be cut by the conductor. The pole coil is also called field coil or exciting coil.

Armature core

The armature of a D.C. machine is in the form of cylindrical form built up of thin lamination of low loss silicon steel as shown in the Figure 6. The lamination are usually of 0.4mm to 0.5mm thickness and are in the form of circular discs as show in Figure 7. These discs or stampings are insulated from one another by a shellac varnish or paper or oxide film coating/ The stampings are held firmly by a key and are mounted on the shaft. For large machines the circular laminations instead of being cut in one piece, are cut into a number of

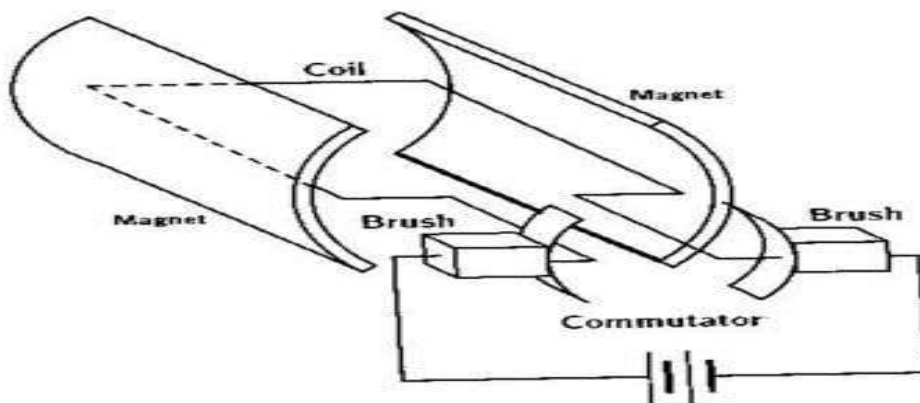
suitable segments as shown in the figure 8 which form part of a complete ring. On the periphery of armature from, slots are provided to accommodate the armature windings. The purpose of armature is the house the armature conductors and to rotate them in the magnetic field, with the help of a prime mover.

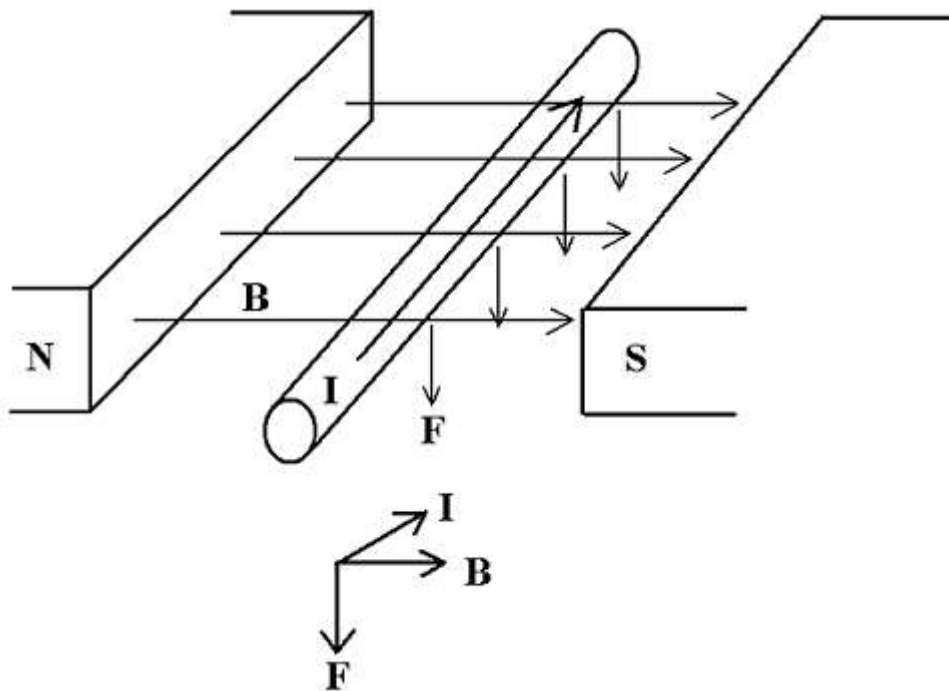
D.C motor

A D.C motor is an electrical machine which converts electrical energy into mechanical energy. Whenever a current carrying conductor is placed in a magnetic field, it experiences a force whose direction is given by Fleming's left hand rule. The working of a D.C. motor is based on this principle.

Principle

Consider a conductor carrying current in a direction away from the observer. The magnetic flux set- up by this current will be in the clockwise direction as shown in the Figure1. The flux lines established by a pair of poles are shown in the Figure 2. Now if the current carrying conductor is placed in the magnetic field established by the two poles, then the resultant of the two fluxes will be such that, the flux lines are weakened on the left side and strengthened on the right side as shown the Figure 3. Since. The force just similar to the force exerted by a elastic rubber band of catapult on the stone piece. If number of such conductors is placed, the force exerted on them will cause rotation.





Constructional Details

A D.C motor construction is similar to that of a D.C. generator. In fact, we can use the same machine as a motor as well as a generator. (For further constructional details and parts refer chapter on D.C generator)

Back EMF

When the motor armature rotates, the conductors housed in the armature also rotate and cut the magnetic lines of force. Hence, an emf is induced in them. This induced emf acts in opposition to the applied voltage and hence it is called back emf 'Eb' whose value is given by

$$E_b = \frac{\phi Z N P}{60 A}$$

This back emf is always less than the applied voltage VL. Thus, we have

$$V_L = E_b + I_a R_a$$

Types of motors

Just similar to D.C.Motors is also classified into three types, based on the manner in which its field winding is connected to its armature.

- (I) Shunt motor
- (ii) Series motor and
- (iii) Compound motor

Q.Explain the Principle of operation of a Transformer.

Or

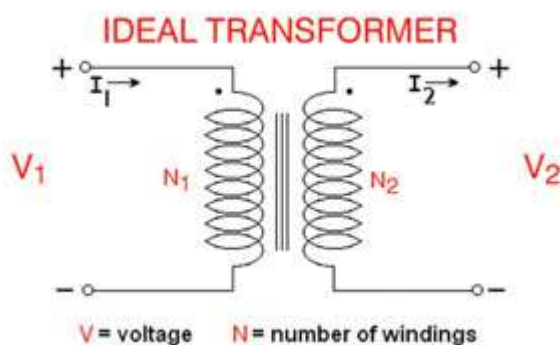
What is Transformer? Draw its constructional view and explain neatly.

TRANSFORMER

The Transformer works on the principle of electromagnetic induction. In this case, the conductors are stationary and the magnetic flux is varying with respect to time. Thus, the induced emf comes under the classification of statically induced emf.

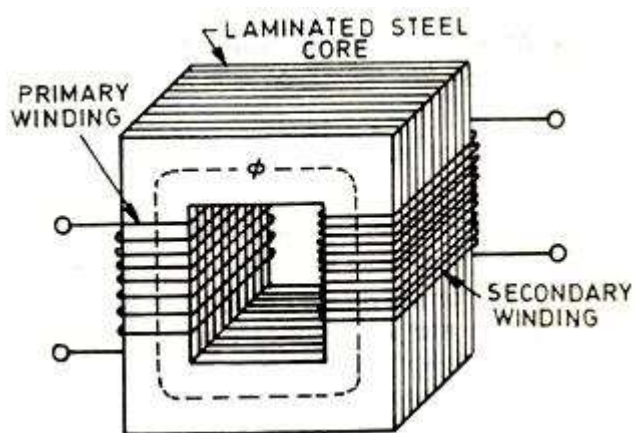
The transformer is a static device of apparatus used to transfer electrical energy from one circuit to another. The two circuits are magnetically coupled. One of the circuits are energized by connecting it to a supply at specific voltage magnitude, frequency and waveform. Then, we have a mutually induced voltage available across the second circuit at the same frequency and waveform but with a change in voltage magnitude as desired.

Transformer is a static electromagnetic device which transforms electric energy from one circuit to another circuit inductively, without change in frequency. It can increase or decrease the voltage level in the circuit with a corresponding decrease or increase in current. When the energy is transformed into a higher voltage, the transformer is called step-up transformer and when the energy is transformed into a lower voltage, the transformer is called step-down transformer.



Principle of operation

Transformer works on mutual induction principle. The two windings of the transformer are magnetically coupled to a core as shown in the Figure 1. When one of the windings called primary is energized with a sinusoidal voltage, an alternating flux is established in both the windings. According to Faraday's laws of electromagnetic induction, an emf is induced in the secondary winding which is called mutually induced emf. This emf induced drives the current through the load connected to the winding. Thus, energy is transferred from primary circuit to the secondary circuit through electromagnetic induction.



Essential requirements of a Transformer are

- i) Magnetic core
- ii) Two windings
- iii) A time varying magnetic flux

The Transformer has N_1 turns in its primary winding and N_2 turns in its secondary winding. The primary winding is connected to a sinusoidal voltage of magnitude V_1 at a frequency ' f ' hertz. A working flux of ϕ webers is set up in the magnetic core. This working flux is alternating and sinusoidal as the applied voltage is alternating and sinusoidal. When

this flux link the primary and the secondary winding, emfs are induced in them. The emf induced in the primary is the self induced emf and that induced in the secondary is the mutually induced emf.

Whenever any load is put on the transformer the primary of the transformer draws the required amount of current in order to keep the working flux constant. Thus the transformer works with a perfect static balance.

Constructional details

The essential parts of a transformed are

- (i) primary and secondary windings and
- (ii) transformer core

Primary winding

The transformer winding to which alternating supply is given is called primary winding.

Secondary winding

The transformer winding to which load is connected (or from which the energy is received) is called secondary winding.

Transformer core

It magnetically couples the two windings of the transformer. The laminated core reduces the eddy current loss. Each lamination is of thickness 0.35mm to 0.5 mm. Hysteresis loss in the core is reduced by using silicon steel as the material for the core. The two basic types of transformer cores are core type and shell type.

Core type transformer

In this type of transformer, the windings surround a considerable part of the core as shown in Figure 2. The coils used are of cylindrical type and are form-wound in helical layers with the different layers insulated from each other by paper, cloth or mica boards. The individual lamination of the core is cut in the form of L shaped long strips as shown in Figure 3.

Shell type transformer

In the type transformer, the core surrounds a considerable part of the windings as shown in the Figure 4. The individual laminations of the core are cut in the form of E or I shaped long strips as shown in the Figure 5. The coils used are Form-wound in the form of pancakes and are of multilayer disc type. Each layer is insulated from one another by means of paper.

Q. Explain the Principle of operation of a Synchronous generator.

Or

Sketch neatly and explain about Synchronous Generator.

SYNCHRONOUS GENERATOR

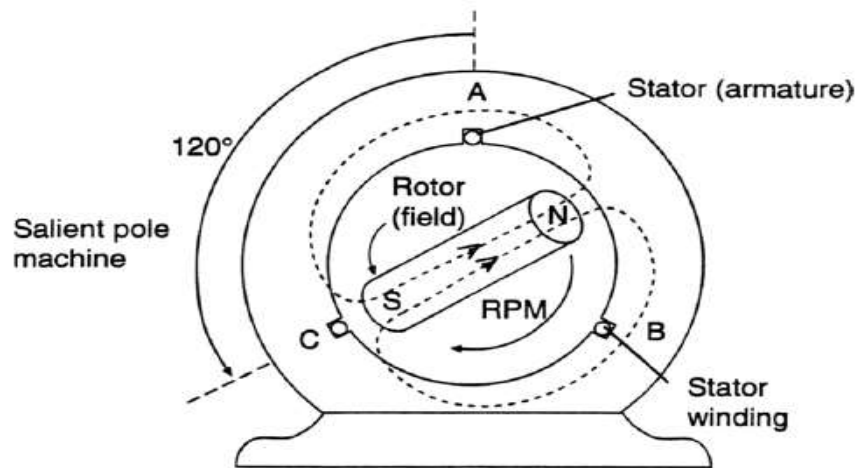
PRINICPLE OF OPERATION:

It works on the principle of electromagnetic induction. If a conductor is placed in a moving magnet field an emf is induced in the stationary conductor as per Faraday's Law of Electromagnetic induction. The two important parts of Synchronous generators are Stator and Rotor.

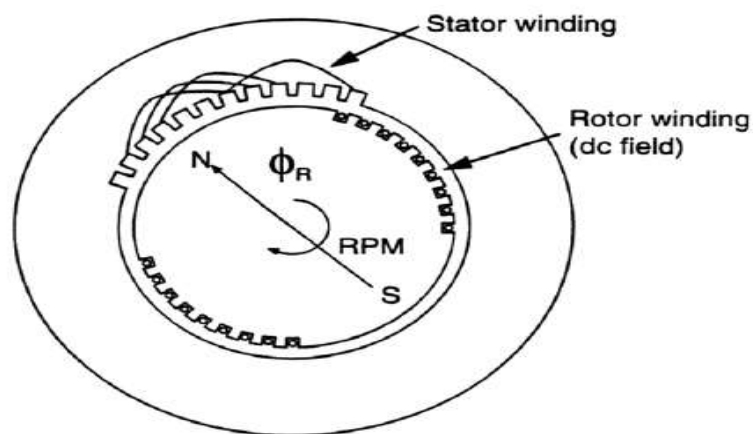
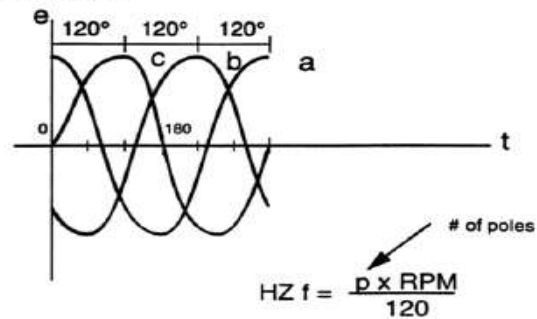
A description of the design criteria leading to the construction of a modern turbo generator, as well as contains a detailed description of all components most commonly found in such a machine. This section is limited to the presentation of the basic components comprising a synchronous machine, with the purpose of describing its basic operating theory. Synchronous machines come in all sizes and shapes, from the miniature permanent magnet synchronous motor in wall-clocks, to the largest steam-turbine driven generators of up to about 1500 MVA.

Synchronous machines are one of two types: the stationary field or the rotating dc magnetic field. The stationary field synchronous machine has salient poles mounted on the stator—the stationary member. The poles are magnetized either by permanent magnets or by a dc current. The armature, normally containing a three-phase winding, is mounted on the shaft. The armature winding is fed through three slip rings (collectors) and a set of brushes sliding on them. This arrangement can be found in machines up to about 5 kVA in rating. The rotating magnetic field (also known as revolving-field) synchronous machine has the field-winding wound on the rotating member (the *rotor*), and the *armature* wound on the stationary member (the *stator*). A dc current, creating a magnetic field that must be rotated at synchronous speed, energizes the rotating field-winding.

The stator consists of a cast iron (or) mild steel frame, which supports the armature core. This frame acts as an enclosure and provides a closed path for the magnetic flux. The armature core is made of laminated sheets. The material for armature core may be special magnetic iron (or) steel alloys. The inner periphery of the armature core is slotted in order to accommodate the armature winding.



Voltage waveforms for phases A, B, C



WORKING:

The field magnets are magnetized by applying 125 volts (or) 250 volts through slip rings. The field windings are connected such that, alternate N and S poles are provided. The rotor and hence the field magnets are driven by the prime movers (steam turbine (or) engine).

As the rotor rotates, the armature conductors are cut by the magnetic flux. Hence emf is induced in the armature conductors. As the magnetic poles are alternatively N and S pole, the emf acts in one direction and then in the other direction. Hence an alternating emf induced in stator conductors. The frequency of induced emf is depends on the number of N and S poles moving past an armature conductor in one second. The frequency of induced emf is given by,

$$f = PN / 120$$

P – No. of poles

N – Speed of rotor in rpm

The direction of induced emf can be obtained by Fleming's Right hand rule.

Q. Explain the Principle of operation of a Induction Motor.

Or

Explain about the operation of a Split phase capacitor start induction motor.

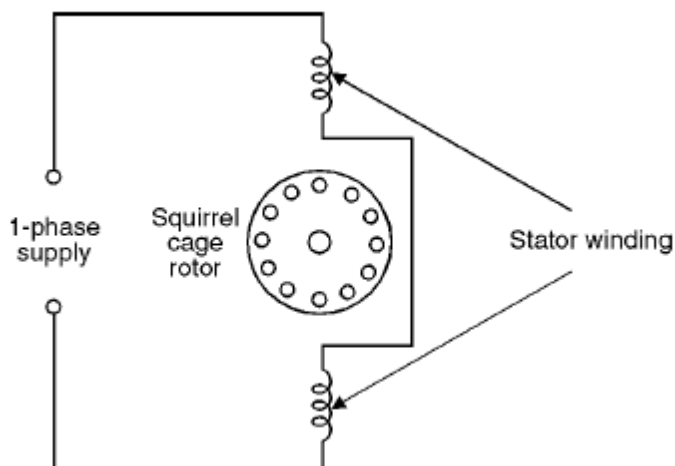
INDUCTION MOTOR

Single phase induction motors are widely used in domestic, industrial and machine tool applications. The capacity of this motor varies from fractional horse power to 5HP.

Fractional horse power motors are used in variety of applications.

Single phase induction motor

Single phase induction motors are used n variety of applications at home, factory, office and business establishments, Based on the method of starting, these motors Are classified into the following types:



(i) Split-phase motors

(ii) Shaded pole motors

The construction of single phase motor is more or less similar to Poly phase induction motor. The difference is that, its stator is wound for single phase. The rotor is similar to a squirrel cage rotor. The main draw back in this motor is that it is not self starting.

Split-phase motors

Since the single phase induction motor is not self-starting, additional arrangement has to be made to make it self-starting. This could be achieved by using two windings with large phase difference between the currents carried by them. This arrangement produced a revolving flux and hence makes the motor self starting. Depending on the circuit element connected in series with the auxiliary or starting winding, the split phase motors are classified into

- (i) Resistance-start induction motor
- (ii) Capacitance-start induction motor
- (iii) Capacitance-start-and-run motor
- (iv) Two-value-capacitor-run motor.

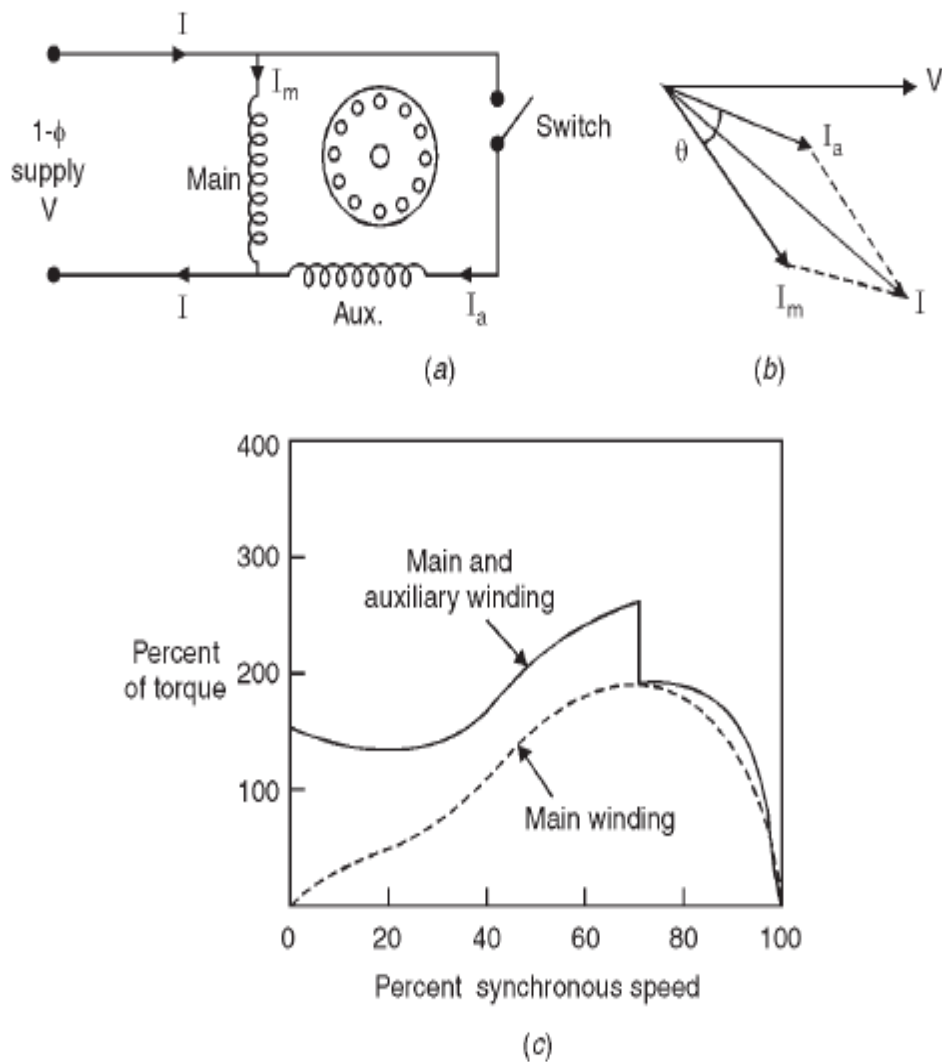


Fig. 9.5. Split phase induction motor (a) Connection (b) Phasor diagram at starting (c) Typical torque-speed characteristic.

CONSTRUCTION:

The construction of 1 ϕ is similar to 3 ϕ Induction motors. The stator has a single winding. The rotor is squirrel cage type as in 3 ϕ Induction motors. When starter winding is energized with single phase supply an alternating flux is setup, but it is not a revolving field as in 3 ϕ induction motor. This alternating flux when acting on stationary motor cannot produce rotation. Only a revolving field results in the rotation of rotor. So single phase induction motor is not self starting. However if the rotor is given an initial start by hand (or) by other means then motor may start and run.

Resistance-start induction motor

The arrangement is for resistance start induction motor as shown in the Figure 8. The starting winding has a high resistance R connected in series with it. The current flowing through it is given by I_s .

The centrifugal switch S disconnects the starting winding when the motor reaches 80% of its full load speed. The main winding has low resistance and high reactance and it carries a current of I_m . The torque developed by the motor is proportional to $\sin\alpha$, where α is the angle between I_m and I_s . Thus for obtaining high torque, α should be as high as possible.

Split phase capacitor start induction run motor:

The single phase supply given to the stator winding, can be split into two phases by connecting a suitable capacitor in series with the starting winding. The value of capacitance should be such that the phase angle α between I_s and I_m should be nearer to 90° . Then the fluxes due to this current is also displaced by the same angle α . This causes revolving field. The starting torque developed is about 300 to 450% of Full load torque. This is more as compared to the previous type motor as α is more here. The value of α in the previous type motor is around 40° only. When the motor attains 75% of rated speed, the centrifugal switch opens and the starting winding is isolated from the circuit. This motor is used when high starting torque is needed under loaded conditions. They are used for pumps, refrigeration units, air-conditioners, large size washing machines etc.

Q.Explain about the concepts of sources of electrical energy conversion.

Or

Write short notes on electrical energy conversion.

SOURCES FOR ELECTRICAL ENERGY CONVERSION:

Generation of Electrical energy is defined as the conversion of energy available in different forms in nature into electrical energy.

Prime mover is coupled with alternator. It rotates the rotor of the alternator. DC supply is given to rotor. The emf is induced in armature conductors. Here mechanical energy (output of prime mover) is converted to electrical energy. (output of alternator).

Sources of energy:

Electrical energy is produced from energy available in various forms in nature. The various sources of energy are:

- i) Sun
- ii) Wind
- iii) Water
- iv) Fuel
- v) Nuclear energy

The energy due to sun and wind has not been used in large, scale due to certain limitations.

Generating stations:

The generating stations are classified into following categories depending upon the form of energy converted:

- i) Thermal power stations
- ii) Hydro electric power stations
- iii) Diesel power station
- iv) Nuclear power station

Thermal power station:

This type of power station converts heat energy, produced from combustion of coal, into electrical energy. A thermal station basically works on Rankin cycle.

Advantages:

- i) The fuel. (ie) coal used in quite cheap.
- ii) Less initial cost compared to other generating stations.
- iii) As coal can be transported to site, it can be installed at any place.
- iv) Compared to hydroelectric station, it needs less space.
- v) Compared to diesel power station, the per unit cost of generation is less.

Disadvantages:

- i) Large amount of smoke and fumes are produced. So it pollutes the system atmosphere.
- ii) It's running cost is higher as compared to hydroelectric plant.

Q.Draw the Schematic Arrangement Of Thermal Power Station and explain neatly

SCHEMATIC ARRANGEMENT OF THERMAL POWER STATION:

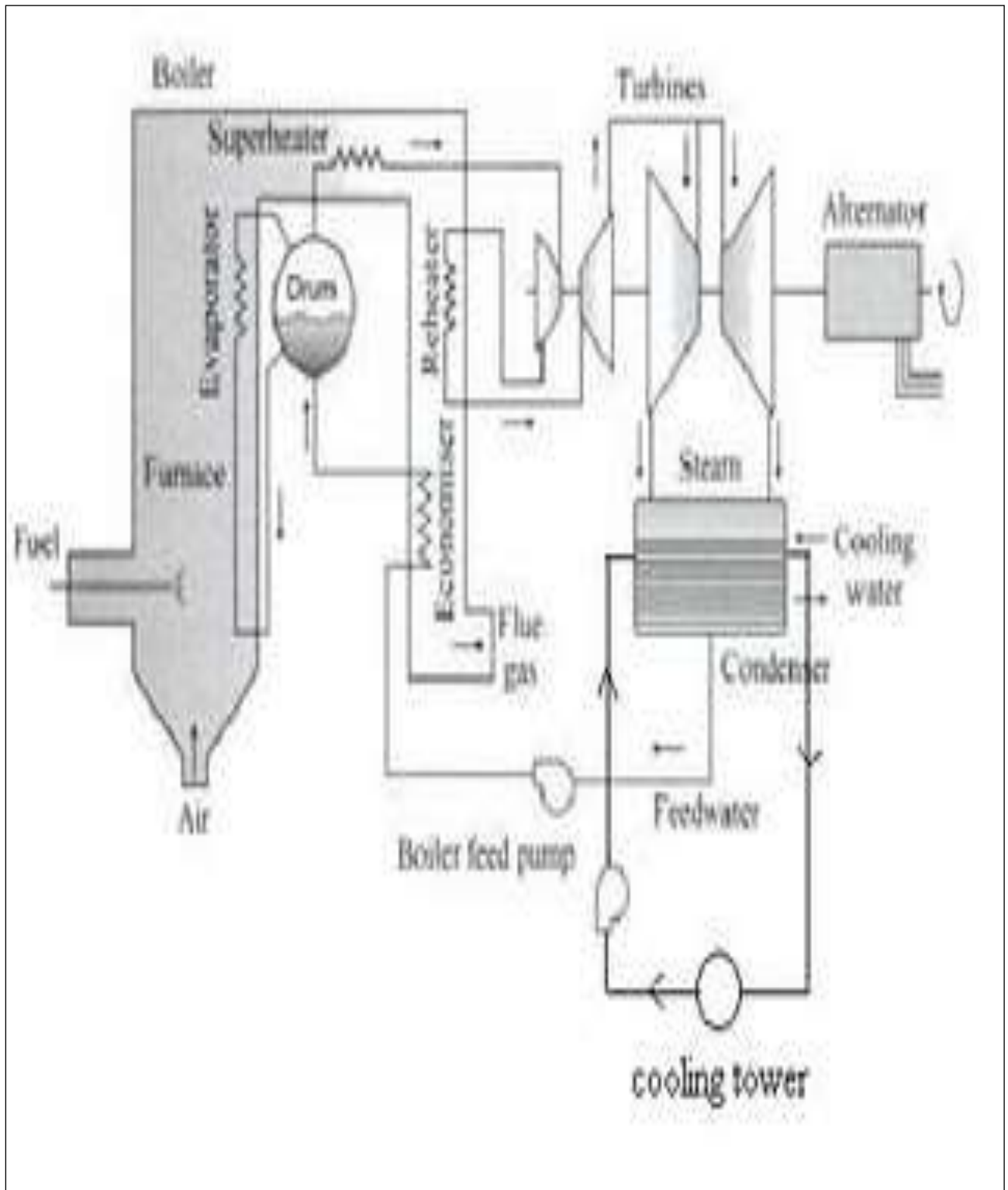
The whole arrangement is divided into the following stages:

- i) Coal and ash handling plant
- ii) Steam generating plant
- iii) Steam turbine
- iv) Alternator
- v) Feed water
- vi) Cooling arrangement

Choice of site for steam power station:

- i) Supply of fuel
- ii) Availability
- iii) Transportation facility
- iv) Cost and type of land
- v) Nearness to load centres
- vi) Distance from populated areas

SCHEMATIC DIAGRAM OF THERMAL POWER STATION:



Main parts of the plant are

1. Coal conveyor
2. Stoker
3. Pulverizer
4. Boiler

5. Coal ash
6. Air preheater
7. Electrostatic precipitator
8. Smoke stack
9. Turbine
10. Condenser
11. Transformers
12. Cooling towers
13. Generator
14. High - voltage power lines

BASIC OPERATION :

A thermal power plant basically works on **Rankine cycle**.

COAL CONVEYOR :

This is a belt type of arrangement. With this coal is transported from coal storage place in power plant to the place near by boiler.

STOKER :

The coal which is brought near by boiler has to put in boiler furnace for combustion. This stoker is a mechanical device for feeding coal to a furnace.

PULVERIZER :

The coal is put in the boiler after pulverization. For this pulverizer is used. A pulverizer is a device for grinding coal for combustion in a furnace in a power plant.

BOILER :

Now that pulverized coal is put in boiler furnace. Boiler is an enclosed vessel in which water is heated and circulated until the water is turned in to steam at the required pressure.

Coal is burned inside the combustion chamber of boiler. The products of combustion are nothing but gases. These gases which are at high temperature vaporize the water inside the boiler to steam. Some times this steam is further heated in a **superheater** as higher the steam pressure and temperature the greater efficiency the engine will have in converting the heat in steam in to mechanical work. This steam at high pressure and tempeture is used directly as a heating medium, or as the working fluid in a prime mover to convert thermal energy to mechanical work, which in turn may be converted to electrical energy. Although other fluids are sometimes used for these purposes, water is by far the most common because of its economy and suitable thermodynamic characteristics.

SUPERHEATER :

Most of the modern boilers are having superheater and reheater arrangement. Superheater is a component of a steam-generating unit in which steam, after it has left the boiler drum, is heated above its saturation temperature. The amount of superheat added to the steam is influenced by the location, arrangement, and amount of superheater surface installed, as well as the rating of the boiler. The superheater may consist of one or more stages of tube banks arranged to effectively transfer heat from the products of combustion. Superheaters are classified as convection, radiant or combination of these.

REHEATER :

Some of the heat of superheated steam is used to rotate the turbine where it loses some of its energy. Reheater is also steam boiler component in which heat is added to this intermediate-pressure steam, which has given up some of its energy in expansion through the high-pressure turbine. The steam after reheating is used to rotate the second steam turbine (see Layout fig) where the heat is converted to mechanical energy. This mechanical energy is used to run the alternator, which is coupled to turbine, thereby generating electrical energy.

CONDENSER :

Steam after rotating steam turbine comes to condenser. Condenser refers here to the shell and tube heat exchanger (or surface condenser) installed at the outlet of every steam turbine in Thermal power stations of utility companies generally. These condensers are heat exchangers which convert steam from its gaseous to its liquid state, also known as phase transition. In so doing, the latent heat of steam is given out inside the condenser. Where water is in short supply an air cooled condenser is often used. An air cooled condenser is however significantly more expensive and cannot achieve as low a steam turbine backpressure (and therefore less efficient) as a surface condenser.

COOLING TOWERS :

The condensate (water) formed in the condenser after condensation is initially at high temperature. This hot water is passed to cooling towers. It is a tower- or building-like device in which atmospheric air (the heat receiver) circulates in direct or indirect contact with warmer water (the heat source) and the water is thereby cooled (see illustration). A cooling tower may serve as the heat sink in a conventional thermodynamic process, such as refrigeration or steam power generation, and when it is convenient or desirable to make final heat rejection to atmospheric air. Water, acting as the heat-transfer fluid, gives up heat to atmospheric air, and thus cooled, is recirculated through the system.

ECONOMISER :

Flue gases coming out of the boiler carry lot of heat. Function of economiser is to recover some of the heat from the heat carried away in the flue gases up the chimney and utilize for heating the feed water to the boiler. It is placed in the passage of flue gases in between the exit from the boiler and the entry to the chimney. The use of economiser results in saving in coal consumption, increase in steaming rate and high boiler efficiency but needs extra investment and increase in maintenance costs and floor area required for the plant. This is used in all modern plants. In this a large number of small diameter thin walled tubes are placed between two headers. Feed water enters the tube through one header and leaves through the other. The flue gases flow out side the tubes usually in counter flow.

AIR PREHEATER :

The remaining heat of flue gases is utilized by air preheater. It is a device used in steam boilers to transfer heat from the flue gases to the combustion air before the air enters the furnace. Also known as air heater; air-heating system. It is not shown in the lay out. But it is kept at a place near by where the air enters in to the boiler. The purpose of the air preheater is to recover the heat from the flue gas from the boiler to improve boiler efficiency by burning warm air which increases combustion efficiency, and reducing useful heat lost from the flue.

GENERATOR :

An alternator is an electromechanical device that converts mechanical energy to alternating current electrical energy. Most alternators use a rotating magnetic field. Different geometries - such as a linear alternator for use with stirling engines - are also occasionally used. In principle, any AC generator can be called an alternator, but usually the word refers to small rotating machines driven by automotive and other internal combustion engines.

TRANSFORMERS :

It is a device that transfers electric energy from one alternating-current circuit to one or more other circuits, either increasing (stepping up) or reducing (stepping down) the voltage. Uses for transformers include reducing the line voltage to operate low-voltage devices (doorbells or toy electric trains) and raising the voltage from electric generators so that electric power can be transmitted over long distances. Transformers act through electromagnetic induction; current in the primary coil induces current in the secondary coil. The secondary voltage is calculated by multiplying the primary voltage by the ratio of the number of turns in the secondary coil to that in the primary.

Q.Draw the Schematic Arrangement Of Hydro electric Power Station and explain neatly.

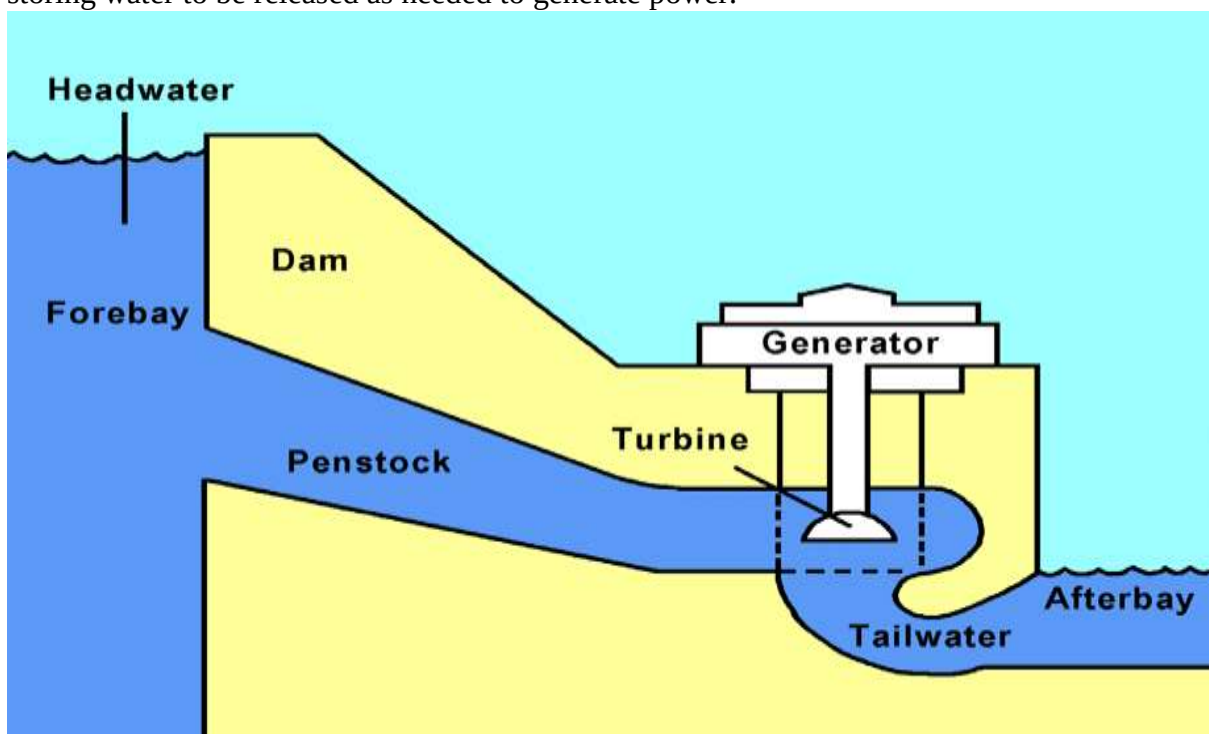
HYDROELECTRIC POWER

It is a form of energy ... a renewable resource. Hydropower provides about 96 percent of the renewable energy. Other renewable resources include geothermal, wave power, tidal power, wind power, and solar power. Hydroelectric power plants do not use up resources to create electricity nor do they pollute the air, land, or water, as other power plants may. Hydroelectric power has played an important part in the development of this Nation's electric power industry. Both small and large hydroelectric power developments were instrumental in the early expansion of the electric power industry.

Hydroelectric power comes from flowing water ... winter and spring runoff from mountain streams and clear lakes. Water, when it is falling by the force of gravity, can be used to turn turbines and generators that produce electricity.

To generate electricity, water must be in motion. This is kinetic (moving) energy. When flowing water turns blades in a turbine, the form is changed to mechanical (machine) energy. The turbine turns the generator rotor which then converts this mechanical energy into another energy form -- electricity. Since water is the initial source of energy, we call this hydroelectric power or hydropower for short. At facilities called hydroelectric power plants, hydropower is generated.

Some power plants are located on rivers, streams, and canals, but for a reliable water supply, dams are needed. Dams store water for later release for such purposes as irrigation, domestic and industrial use, and power generation. The reservoir acts much like a battery, storing water to be released as needed to generate power.



The dam creates a head or height from which water flows. A pipe (penstock) carries the water from the reservoir to the turbine. The fast-moving water pushes the turbine blades, something like a pinwheel in the wind. The water's force on the turbine blades turns the rotor, the moving part of the electric generator. When coils of wire on the rotor sweep past the generator's stationary coil (stator), electricity is produced.

Once the electricity is produced, it must be delivered to where it is needed -- our homes, schools, offices, factories, etc. Dams are often in remote locations and power must be transmitted over some distance to its users.

Q.Explain about Main components of the power systems.

Or

What are the main components in transmission and Distribution systems and explain neatly?

Main components of transmission and distribution system:

Various components of power supply system:

- 1. Generating station**
- 2. Primary transmission**
- 3. Secondary transmission**
- 4. Primary distribution**
- 5. Secondary distribution**
- 6. Feeders**
- 7. Distributors**
- 8. Service mains**

POWER GENERATION:

Electrical power is generated normally at 11KV (or) at 6.6 KV by number of generators in parallel. Generation at still higher voltage will improve some technical problems. Power generated at 11 KV is stepped up to 110 KV (or) 230 KV (or) 400 KV (or) still higher voltage with the help of power transformers.

TRANSMISSION SYSTEM:

The power generated at the generating stations is transmitted at higher voltage to the main load centre. This transmission system is known as primary transmission system. The voltage level here may be 1500KV, 700KV, 400KV (or) 230KV. There are advantages when power is transmitted at high voltage.

- i) The volume of conductor material required is low.

- ii) For a given amount of power transmitted, the current through the line is reduced, when the transmission system voltage is high.
- iii) As the current is reduced the voltage in the line is reduced and hence the line regulation is improved.

Advantages of DC Transmission over AC Transmission:

- i) The amount of insulation required is less
- ii) The size of towers, cross arms required are small
- iii) It improves the stability of the system.
- iv) Power factor of DC Transmission is unity.
- v) There is no charging current

Distribution system:

The power is received at the sub load center at voltages of 33KV, 66KV (or) 110KV. The voltage level is stepped down to 11KV at the sub load centers. The power is distributed through the 11KV lines, which runs along the main roads. This distribution system is referred to as primary distribution system.

8.

Main components of transmission and distribution system

Conductors, ---Which are used to carry the electrical power from one place to another place.

Supports --Which may be RCC poles, MS tubular poles (or) towers. These are used to keep the conductors at a proper height from the ground.

Cross arms --Which are attached to the poles (or) towers and are used to support the conductors.

Insulators --Which are used to isolate the conductors at higher potential from the ground.

Lighting Arrester – To safe guard from lightening

Ground & Stay wires: - Earthing and supporting purpose.

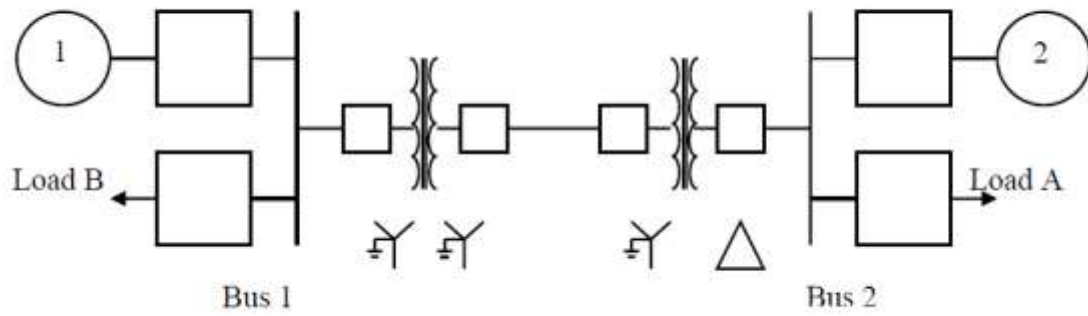
Q. What is one line diagram and explain with neat sketch.

ONE-LINE DIAGRAM or SINGLE LINE DIAGRAM

The transmission of electric power from power station to consumer's premises is known as electric power system. A one-line diagram is a simplified graphical representation of a three phase power system, used extensively in power flow studies. In power engineering, one can make the assumption that the three phases of a system are balanced and can therefore be examined as a single phase. The assumption can be made because what happens on one of the three balanced phases, theoretically, will happen to all three phases. This makes the evaluation of the system much less complicated without losing any information. Per unit voltage used extensively in one-line diagrams to further simplify the process.

The main components of a one-line (or single line) diagram are; Buses, Branches, Loads, Machines, 2 Winding Transformers, Switched Shunts, Reactor and Capacitor Banks.

Generated voltage = 11kv. This voltage is stepped up to 132 KV by a step up transformer for primary distribution to reduce the transmission loss. After primary transmission, this voltage (132 KV) is stepped down by a step down transformer for secondary transmission. This is further stepped down to 11KV before from substation primary distribution is carried out. Further this is stepped down to 440 V for secondary distribution. Consumers supply is given by three phase 4 wire line.



A one-line diagram of a simple power system

Oil or liquid circuit breaker

Rotating machine

Two-winding power transformer

Wye connection, neutral ground